



Linear Canonical Bargmann Transform

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Abstract

In this paper, a linear canonical transform associated with the Bargmann transform, referred to as the linear canonical Bargmann transform (LCBT) is proposed. The relationship between the Fourier transform, fractional Fourier transform, and the LCBT are discussed. Following this, the basic properties of the LCBT are derived, including the Parseval theorem, linearity, translation, modulation, convolution, and the uncertainty principle. It is evident that the LCBT serves as a generalized form of both the Fourier transform and fractional Fourier transform.

Keywords Linear canonical Bargmann transform · Bargmann transform · Convolution · Uncertainty principle

1 Introduction

The Bargmann transform B is the operator, which is a concept in mathematical analysis and mathematical physics. It was first introduced in [1] and further studied in [2]. The Bargmann transform can be regarded as a bridge between real functions $f(x)$ and complex functions $f(z)$, which is an integral operator from $L^2(\mathbb{R}) \rightarrow F^2(\mathbb{C})$, defined by

$$B[f(x)](z) = q \int_{\mathbb{R}} f(x) e^{2xz - x^2 - \frac{1}{2}z^2} dx, \quad (1)$$

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where $q = (\frac{2}{\pi})^{\frac{1}{4}}$, the inverse of the B is also an integral operator from $F^2(\mathbb{C}) \rightarrow L^2(\mathbb{R})$, namely

$$B^{-1}[f(z)](x) = q \int_{\mathbb{C}} f(z) e^{2x\bar{z} - x^2 - \frac{1}{2}\bar{z}^2} d\lambda(z), \quad (2)$$

where $F^2(\mathbb{C})$ is the Fock space F^2 (FSF).

The FSF is the Hilbert space of all entire functions $f(z)$, and plays a very important role in the fields of quantum information, harmonic analysis, and partial differential equations [3–6], $f(z)$ meets the following conditions

$$\|f\|^2 = \int_{\mathbb{C}} |f(z)|^2 d\lambda(z) < \infty, \quad (3)$$

and

$$f(z) = \int_{\mathbb{C}} f(w) e^{z\bar{w}} d\lambda(w), \quad (4)$$

where $d\lambda(z) = \frac{1}{\pi} e^{-|z|^2} dA(z)$, $dA(z) = dx dy$ is the Lebesgue measure, $f(z)$ is called the reproducing formula for functions in the Fock space, $f(z) \in F^2(\mathbb{C})$, $z \in \mathbb{C}$.

To further explore the theory and applications of the FSF, many experts have employed the Bargmann transform as a important tool [7–12]. After 1967, more in depth researches related to the Bargmann transform are carried out. For instance, Shih uses the truncation method to derive the closed form of the Bargmann transform for white noise functionals associated with the Lévy spectrum [8]. Additionally, Dong et al. study the impact of the Bargmann transform on several classic integral operators, including the Fourier transform (FT), Hilbert transform (HT), and wavelet transform (WT) [9, 12]. Toft focuses on the function images and distribution spaces under the Bargmann transform [10], while Cao et al. study its mapping properties on $L^p(\mathbb{R})$ spaces of the real line [11]. Furthermore, Zhu proposes a new integral operator based on Bargmann transform, and deduces some new forms of classical operators in the FSF [6].

The above researchers mostly combine the Bargmann transform with classical integral transforms (FT, HT, WT), and important conclusions are obtained in the FSF. However, It is well known that linear canonical transform (LCT) can include the FT [13], fractional Fourier transform (FRFT) [14], Fresnel transform (FRT) [15] and Lorenz transform (LRT) [16] under different specific conditions. As a linear integral transform with three free parameters, the LCT provides flexibility and versatility, allowing it to adapt to various application scenarios [17]. Therefore, the combination of the Bargmann transform with the LCT is expected to further expand the mathematical theoretical framework of integral transforms in the FSF and is regarded as a research hotspot in the fields of mathematics and information analysis [13–17]. Given a real parameter matrix $A = (a, b, c, d)$, and the parameters satisfies the condition $ad - bc = 1$, the LCT of any $f(x) \in L^2(\mathbb{R})$ with respect to the matrix A is defined by

$$L_A(f(x))(u) = \int_{\mathbb{R}} f(x) K_A(x, u) dx, \quad (5)$$

and the $K_A(x, u)$ is given by

$$K_A(x, u) = \begin{cases} \frac{1}{\sqrt{i2\pi b}} e^{i \frac{ax^2 + du^2 - 2xu}{2b}}, & b \neq 0, \\ \sqrt{d} e^{i \frac{cd u^2}{2}} \delta(x - du), & b = 0, \end{cases}$$

where L_A is the LCT, δ is the Dirac delta function.

In the $L^2(\mathbb{R})$ space, the theoretical framework and fundamental properties of the LCT have undergone extensive research and development. The establishment of important properties, such as the convolution theorem, have provided a solid theoretical foundation for the widespread application of the LCT in practical settings [17]. These properties reveal the unique advantages and flexibility of the LCT in signal processing, making it an indispensable analytical tool in fields such as optics, quantum technologies, marine science, aerospace, biomedicine, and communications [18, 19]. However, with the rapid development of emerging technologies such as quantum computing and optical communication, the theoretical knowledge of the LCT in $L^2(\mathbb{R})$ space is not sufficient to address the new challenges posed by these cutting edge fields [4, 19]. In this context, this paper aims to explore the definition and fundamental properties of the Linear canonical Bargmann transform (LCBT) related to the FSF. By applying Parseval's theorem, linearity, translation, modulation, convolution, and the uncertainty principle of the LCBT, researchers are enabled to gain a deeper understanding of the mathematical representation of signals. The LCBT combines the classical properties of the LCT with the analytical structure of the Bargmann transform, providing a powerful mathematical framework for applications in optical communication and quantum technologies [4, 15, 20].

The remainder of this paper is structured as follows. In Sect. 1, basic theories, such as the definition of the LCT, FSF, and the Bargmann transform is briefly reviewed. In Sect. 2, the definition and basic properties of the LCBT are proposed, and the relationship between the FT, FRFT, and the LCBT in the FSF are discussed. In Sect. 3, the convolution and uncertainty principle in the FSF is derived. In Sect. 4, this paper is summarized, and future research directions are pointed out.

2 Definition and Properties

This section delves into the conceptual framework of the LCBT, meticulously defining its essence and exploring its fundamental properties. We aim to establish a clear understanding of the LCBT's intrinsic nature, elucidating its intricate relationship with both the conventional FT and FRFT within the context of the FSF. This exploration seeks to unveil the LCBT's potential for advancing our comprehension of signal and image processing [10, 13, 14].

Definition 1 (LCBT) Let the LCBT parameter matrix is $A = (a, b, c, d)$, and $ad - bc = 1$, the LCBT is defined as

$$T = BL_A B^{-1}, \quad (6)$$

consequently, the inverse operator

$$T^{-1} = BL_A^{-1} B^{-1}, \quad (7)$$

where L_A is the LCT, L_A^{-1} is the inverse operator of the L_A , T^{-1} is the inverse operator of the T , B is the Bargmann transform in equations (1), B^{-1} is the inverse operator of the B .

Properties 1 Let $f(w) \in F^2(\mathbb{C})$, assuming the LCBT parameter is $A = (a, b, -b, a)$, and $a^2 + b^2 = 1$, according to Definition 1, then

$$T[f(w)](z) = \sqrt{\frac{1}{a+ib}} \int_{\mathbb{C}} f(w) e^{\frac{1}{a+ib} z \bar{w}} d\lambda(w), \quad (8)$$

by Equation 4, it can be found that

$$T[f(w)](z) = \sqrt{\frac{1}{a+ib}} f\left(\frac{1}{a+ib} z\right).$$

The specific form of the transform, particularly the use of complex variables and exponential functions, suggests applications in areas involving wave mechanics, signal processing, or optics, where the analysis of phase and amplitude plays a crucial role. The factor $\sqrt{\frac{1}{a+ib}}$ and the integral suggest that this is a normalized, possibly unitary transform, preserving certain properties like total probability or energy.

Properties 2 : Let the F is the FT in $L^2(\mathbb{R})$ space, $f(w) \in F^2(\mathbb{C})$, assuming the LCBT parameter is $A = (0, 1, -1, 0)$, then

$$BL_A B^{-1}[f(w)](z) = \sqrt{-i} B F B^{-1}[f(w)](z) = \sqrt{-i} f(-iz). \quad (9)$$

Proof Let the parameter $A = (0, 1, -1, 0)$ in Properties 1, then

$$\begin{aligned} BL_A B^{-1}[f(w)](z) &= \sqrt{\frac{1}{i}} \int_{\mathbb{C}} f(w) e^{\frac{1}{i} z \bar{w}} d\lambda(w) \\ &= \sqrt{-i} \int_{\mathbb{C}} f(w) e^{-iz \bar{w}} d\lambda(w) \\ &= \sqrt{-i} f(-iz), \end{aligned}$$

further simplification using the Gaussian integral over t gives

$$\begin{aligned}
 F(B^{-1}f)(x) &= q \sqrt{\frac{1}{\pi}} \int_{\mathbb{R}} e^{-2ixt} dt \int_{\mathbb{C}} f(z) e^{2t\bar{z} - t^2 - \frac{1}{2}\bar{z}^2} d\lambda(z) \\
 &= q \sqrt{\frac{1}{\pi}} \int_{\mathbb{C}} f(z) e^{-\frac{1}{2}\bar{z}^2} d\lambda(z) \int_{\mathbb{R}} (e^{-t^2 + 2t(\bar{z} - ix)}) dt \\
 &= q e^{-x^2} \int_{\mathbb{C}} f(z) e^{-\frac{1}{2}\bar{w}^2 - 2ix\bar{w}} d\lambda(z),
 \end{aligned}$$

finally, we arrive at the result

$$\begin{aligned}
 (BFB^{-1}f)(z) &= q^2 e^{-\frac{1}{2}z^2} \int_{\mathbb{R}} e^{2xz - 2x^2} dx \int_{\mathbb{C}} f(w) e^{\frac{1}{2}\bar{w}^2 - 2ix\bar{w}} d\lambda(w) \\
 &= q^2 e^{-\frac{1}{2}z^2} \int_{\mathbb{C}} f(w) e^{\frac{1}{2}\bar{w}^2} d\lambda(w) \int_{\mathbb{R}} e^{-2x^2 + 2x(z - i\bar{w})} dx \\
 &= \int_{\mathbb{C}} f(w) e^{-iz\bar{w}} d\lambda(w) \\
 &= f(-iz),
 \end{aligned}$$

the proof is also provided in detail in [6].

However, we found that this is no exception in the FSF. When the parameter $A = (0, 1, -1, 0)$ is present, the LCBT not only elegantly degenerates into the FT in the FSF, accompanied by a unique phase factor $\sqrt{-i}$, but also demonstrates its extraordinary universality in the more complex space [13, 14]. This discovery reveals the profound connections of the LCBT in different mathematical spaces, also provides us with a new perspective to examine and understand the widespread application and profound impact of this transform in fields such as signal processing and image analysis.

Properties 3 : Let the F_α is the FRFT in $L^2(\mathbb{R})$ space, $f(w) \in F^2(\mathbb{C})$, assuming the LCBT parameter is $A = (\cos \alpha, \sin \alpha, -\sin \alpha, \cos \alpha)$, then

$$BL_A B^{-1}[f(w)](z) = \sqrt{e^{-i\alpha}} B F_\alpha B^{-1}[f(w)](z) = \sqrt{e^{-i\alpha}} f(e^{-i\alpha} z). \quad (10)$$

Proof Let the parameter $A = (\cos \alpha, \sin \alpha, -\sin \alpha, \cos \alpha)$ in Properties 1, then

$$\begin{aligned}
 T[f(w)](z) &= \sqrt{e^{-i\alpha}} \int_{\mathbb{C}} f(w) e^{e^{-i\alpha} z \bar{w}} d\lambda(w) \\
 &= \sqrt{e^{-i\alpha}} f(e^{-i\alpha} z),
 \end{aligned}$$

to analyze the expression $F_\alpha(B^{-1}f)(x)$, we start with the following form, we have

$$\begin{aligned}
F_\alpha(B^{-1}f)(x) &= q\sqrt{\frac{1-i\cot\alpha}{2\pi}}e^{ix^2\cot\alpha}\int_{\mathbb{C}}f(z)e^{-\frac{1}{2}z^2}d\lambda(z) \\
&\quad \int_{\mathbb{R}}e^{-(1-i\cot\alpha)t^2+2(\bar{z}-ix\csc\alpha)t}dt \\
&= q\sqrt{\frac{1-i\cot\alpha}{2\pi}}e^{ix^2\cot\alpha}\int_{\mathbb{C}}f(z)e^{-\frac{1}{2}z^2+\frac{(\bar{z}-ix\csc\alpha)^2}{1-icot\alpha}}d\lambda(z) \\
&\quad \int_{\mathbb{R}}e^{-(1-i\cot\alpha)(t-\frac{\bar{z}-ix\csc\alpha}{1-icot\alpha})^2}dt \\
&= qe^{ix^2\cot\alpha}\int_{\mathbb{C}}f(z)e^{-\frac{1}{2}z^2+\frac{(\bar{z}-ix\csc\alpha)^2}{1-icot\alpha}}d\lambda(z) \\
&= qe^{(i\cot\alpha-\frac{\csc^2\alpha}{1-icot\alpha})x^2}\int_{\mathbb{C}}f(w)e^{\frac{1}{2}\frac{1+i\cot\alpha}{1-i\cot\alpha}\bar{w}^2-\frac{2i\csc\alpha}{1-icot\alpha}x\bar{w}}d\lambda(w),
\end{aligned}$$

due to $(i\cot\alpha - \frac{\csc^2\alpha}{1-icot\alpha}) = -1$, $(\frac{1+i\cot\alpha}{1-i\cot\alpha}) = -e^{-2i\alpha}$, $(-\frac{2i\csc\alpha}{1-icot\alpha}) = 2e^{-i\alpha}$, after evaluating the resulting Gaussian integral, we have

$$F_\alpha(B^{-1}f)(x) = qe^{-x^2}\int_{\mathbb{C}}f(w)e^{-\frac{1}{2}e^{-2i\alpha}\bar{w}^2+2e^{-i\alpha}x\bar{w}}d\lambda(w),$$

next, we apply the Bargmann transform B to $F_\alpha(B^{-1}f)(x)$,

$$\begin{aligned}
BF_\alpha(B^{-1}f)(x) &= q^2\int_{\mathbb{R}}e^{-2x^2+2xz-\frac{1}{2}z^2}dt\int_{\mathbb{C}}f(w)e^{-\frac{1}{2}e^{-2i\alpha}\bar{w}^2+2e^{-i\alpha}x\bar{w}}d\lambda(w) \\
&= q^2e^{-\frac{1}{2}z^2}\int_{\mathbb{C}}f(w)e^{-\frac{1}{2}e^{-2i\alpha}\bar{w}^2}d\lambda(w)\int_{\mathbb{R}}e^{-2x^2+2x(z+e^{-i\alpha}\bar{w})} \\
&= q^2e^{-\frac{1}{2}z^2}\int_{\mathbb{C}}f(w)e^{-\frac{1}{2}e^{-2i\alpha}\bar{w}^2+\frac{(z+e^{-2i\alpha}\bar{w})^2}{2}}d\lambda(w) \\
&\quad \int_{\mathbb{R}}e^{-2(x-\frac{(z+e^{-2i\alpha}\bar{w})}{2})^2}d(x),
\end{aligned}$$

finally, we find that

$$\begin{aligned}
BF_\alpha(B^{-1}f)(x) &= \int_{\mathbb{C}}f(w)e^{e^{-i\alpha}z\bar{w}}d\lambda(w) \\
&= f(e^{-i\alpha}z).
\end{aligned}$$

Particular, when parameter $A = (\cos\alpha, \sin\alpha, -\sin\alpha, \cos\alpha)$ is used, the LCT degenerates into the FRFT multiplied by a fixed phase factor $\sqrt{e^{-i\alpha}}$ in the FSF.

Theorem 1 (Parseval theorem) *If $f(w) \leftrightarrow Tf[(w)](z)$, $z, w \in \mathbb{C}$, for all $f(w) \in F^2(\mathbb{C})$, then*

$$\int_{\mathbb{C}}\{f(w)\}^2d\lambda(w) = \int_{\mathbb{C}}\{Tf[(w)](z)\}^2d\lambda(z). \quad (11)$$

According to Properties 1, operator T have distance preserving properties. Therefore, the T operator is a unitary operator.

Properties 4 (Linearity) *If two functions $f_1(w)$ and $f_2(w)$ have T transforms $T[f_1(w)]$ and $T[f_2(w)]$ respectively, for any complex numbers m_1 and m_2 , $f(w) = m_1 f_1(w) + m_2 f_2(w)$, then*

$$T[f(w)] = m_1 T[f_1(w)] + m_2 T[f_2(w)], \quad (12)$$

the corollary is obvious, and the specific proof process will not be provided here.

Properties 5 (Translation) *If functions $f_1(w)$ have T transform $T[f_1(w)]$, $w_0 \in \mathbb{C}$, $f(w) = f_1(w - w_0)$, then*

$$T[f(w)] = e^{\sqrt{m}z\overline{w_0}} T[f_1(w)], \quad (13)$$

where $m = \frac{1}{a+ib}$.

Proof The proof is simple, then

$$\begin{aligned} T[f(w)] &= T[f_1(w - w_0)] \\ &= \sqrt{m} \int_{\mathbb{C}} f_1(w - w_0) e^{mz\overline{w}} d\lambda(w) \\ &= \sqrt{m} \int_{\mathbb{C}} f_1(w) e^{mz(\overline{w} - \overline{w_0})} d\lambda(w - w_0) \\ &= e^{-mz\overline{w_0}} \sqrt{m} \int_{\mathbb{C}} f_1(w) e^{mz\overline{w}} d\lambda(w) \\ &= e^{-mz\overline{w_0}} T[f_1(w)]. \end{aligned}$$

Properties 6 (Modulation) *If functions $f_1(w)$ have T transform $T[f_1(w)]$, $w_0 \in \mathbb{C}$, $f(w) = e^{mzw_0} f_1(w)$, then*

$$T[f(w)] = T[f_1(w - w_0)]. \quad (14)$$

Proof Prove similarity between thought and Properties 5, then

$$\begin{aligned} T[f(w)] &= T[e^{mzw_0} f_1(w)] \\ &= \sqrt{m} \int_{\mathbb{C}} f_1(w) e^{mz(\overline{w} + \overline{w_0})} d\lambda(w) \\ &= \sqrt{m} \int_{\mathbb{C}} f_1(w - w_0) e^{mz\overline{w}} d\lambda(w) \\ &= T[f_1(w - w_0)]. \end{aligned}$$

Modulation is one of the key technologies for achieving wireless communication, which can convert low frequency signals into high frequency signals for transmission, thus achieving long distance communication.

Theorem 2 Let $f(w) \in F^2(\mathbb{C})$, when the LCBT parameter is $A = (a, b, c, d)$, $ad - bc = 1$, and $a \neq d, b \neq -c$, according to Definition 1, then

$$T[f(w)](z) = \sqrt{\frac{1}{\eta i}} \int_{\mathbb{C}} f(w) e^{i \frac{\sigma \bar{w}^2 + \delta z^2 - 2\bar{w}z}{2\eta}} d\lambda(w), \quad (15)$$

where $\bar{w}$ is the conjugate complex of w , and $\sigma = i \frac{(2b+ia)[(c-4b)+2(a+d)i]+4}{4(2b-ai)}$, $\eta = \frac{(4b-c)-2(a+d)i}{4}$, $\gamma = -\frac{(2b+ai)[(4b+c)+2(d-a)i]+4}{4(2b-ai)}$, $\delta = -i \frac{(4b+c)+2(d-a)i}{4}$.

Proof We assume that f is any polynomial (recall that the polynomials are dense in $F^2(\mathbb{C})$, and under the B^{-1} , they become the Hermite polynomials times the Gauss function, which have very good integrability on the real line), we have

$$\begin{aligned} L_A(B^{-1}f)(x) &= q \frac{1}{\sqrt{i2\pi b}} e^{\frac{id}{2b}x^2} \int_{\mathbb{R}} e^{(\frac{ia}{2b}t^2 - \frac{i2xt}{2b})} dt \int_{\mathbb{C}} f(z) e^{2t\bar{z} - t^2 - \frac{1}{2}\bar{z}^2} d\lambda(z) \\ &= q \frac{1}{\sqrt{i2\pi b}} e^{\frac{id}{2b}x^2} \int_{\mathbb{C}} f(z) e^{-\frac{1}{2}\bar{z}^2} d\lambda(z) \int_{\mathbb{R}} \left(e^{\frac{ia}{2b}t^2 - t^2 + 2t\bar{z} - \frac{i2xt}{2b}} \right) dt \\ &= q \frac{1}{\sqrt{i2\pi b}} e^{\frac{id}{2b}x^2} \int_{\mathbb{C}} f(z) e^{-\frac{1}{2}\bar{z}^2} d\lambda(z) \int_{\mathbb{R}} \left(e^{\left(\frac{ia-2b}{2b}\right)t^2 + 2\left(\frac{2b\bar{z}-ix}{2b}\right)t} \right) dt \\ &= q \frac{1}{\sqrt{i2\pi b}} e^{\frac{id}{2b}x^2} \int_{\mathbb{C}} f(z) e^{-\frac{1}{2}\bar{z}^2 + \frac{(2b\bar{z}-ix)^2}{2b(2b-ia)}} d\lambda(z) \\ &\quad \int_{\mathbb{R}} \left(e^{-(a+ib)(t - \frac{(2b\bar{z}-ix)}{2b(2b-ia)})^2} \right) dt, \end{aligned}$$

further simplification using the Gaussian integral over t gives

$$\begin{aligned} L_A(B^{-1}f)(x) &= q \sqrt{\frac{1}{a+i2b}} e^{\frac{id}{2b}x^2} \int_{\mathbb{C}} f(z) e^{-\frac{1}{2}\bar{z}^2 + \frac{(2b\bar{z}-ix)^2}{2b(2b-ia)}} d\lambda(z) \\ &= q \sqrt{\frac{1}{a+i2b}} e^{\frac{(c+2id)}{2(2b-ia)}x^2} \int_{\mathbb{C}} f(z) e^{\frac{(2b+ia)}{2(2b-ia)}\bar{z}^2 - \frac{2i}{(2b-ia)}x\bar{z}} d\lambda(z), \end{aligned}$$

next, we will replace z in the above equation with w , we have

$$L_A(B^{-1}f)(x) = q \sqrt{\frac{1}{a+i2b}} e^{\frac{(c+2id)}{2(2b-ia)}x^2} \int_{\mathbb{C}} f(w) e^{\frac{(2b+ia)}{2(2b-ia)}\bar{w}^2 - \frac{2i}{(2b-ia)}x\bar{w}} d\lambda(w),$$

to make the expression more concise, we set $E = 2b - ia$, $H = c + 2id$, after simplification, the expression above becomes

$$L_A(B^{-1}f)(x) = q \sqrt{\frac{1}{iE}} e^{\frac{H}{2E}x^2} \int_{\mathbb{C}} f(w) e^{\frac{\bar{E}}{2E}\bar{w}^2 - \frac{2i}{E}x\bar{w}} d\lambda(w),$$

therefore, we have

$$\begin{aligned}
 (BL_A B^{-1} f)(z) &= q^2 \sqrt{\frac{1}{iE}} \int_{\mathbb{R}} e^{2xz - x^2 - \frac{1}{2}z^2 + \frac{H}{2E}x^2 - \frac{2i}{E}x\bar{w}} dx \\
 &\quad \int_{\mathbb{C}} f(w) e^{\frac{\bar{E}}{2E}\bar{w}^2} d\lambda(w) \\
 &= q^2 \sqrt{\frac{1}{iE}} e^{-\frac{1}{2}z^2} \int_{\mathbb{C}} f(w) e^{\frac{\bar{E}}{2E}\bar{w}^2} d\lambda(w) \\
 &\quad \int_{\mathbb{R}} e^{2xz + \left(\frac{H-2E}{2E}\right)x^2 + 2\left(\frac{Ez-i\bar{w}}{E}\right)x} dx \\
 &= q^2 \sqrt{\frac{1}{iE}} e^{-\frac{1}{2}z^2} \int_{\mathbb{C}} f(w) e^{\frac{\bar{E}}{2E}\bar{w}^2 - \frac{2(Ez-i\bar{w})^2}{E(H-2E)}} d\lambda(w) \\
 &\quad \int_{\mathbb{R}} e^{\frac{H-2E}{2E}\left[x + \frac{2(Ez-i\bar{w})}{(H-2E)}\right]^2} dx,
 \end{aligned}$$

finally, we arrive at the result

$$\begin{aligned}
 T[f(w)](z) &= BL_A B^{-1}[f(w)](z) \\
 &= \sqrt{\frac{4i}{H-2E}} \int_{\mathbb{C}} f(w) e^{\frac{\bar{E}(H-2E)+4}{2E(H-2E)}\bar{w}^2 + \frac{4i}{H-2E}z\bar{w} - \frac{1}{2}\left(\frac{H+2E}{H-2E}\right)z^2} d\lambda(w).
 \end{aligned}$$

Next, we set $\sigma = i\frac{\bar{E}(H-2E)+4}{4E}$, $\eta = \frac{2E-H}{4} = i\frac{(2b+ia)[(c-4b)+2(a+d)i]+4}{4(2b-ai)}$, $\beta = \frac{2E-H}{4} = \frac{(4b-c)-2(a+d)i}{4}$, $\gamma = -\frac{\bar{E}(H+2E)+4}{4E} = -\frac{(2b+ai)[(4b+c)+2(d-a)i]+4}{4(2b-ai)}$, $\delta = -i\frac{H+2E}{4} = -i\frac{(4b+c)+2(d-a)i}{4}$, $ad-bc=1$, $\bar{w}$ is the conjugate complex of w , we find that operator T have a surprisingly simple form in the FSF, then

$$T[f(w)](z) = \sqrt{\frac{1}{\beta i}} \int_{\mathbb{C}} f(w) e^{i\frac{\alpha\bar{w}^2 + \delta z^2 - 2\bar{w}z}{2\eta}} d\lambda(w).$$

Through the derivation of Theorem 3, we have successfully transform the form of the LCT in $L^2(\mathbb{R})$ space into an equivalent expression in the FSF. This significant conversion not only deepens our understanding of the essence of the LCT, but also opens up new avenues for exploring its connections with the Fock space.

3 Convolution and Uncertainty Principle

In this section, we will briefly introduce the convolution and uncertainty principles of the LCBT.

Theorem 3 (Convolution) *If two functions $f_1(w)$ and $f_2(w)$ have T transforms $T[f_1(w)]$ and $T[f_2(w)]$ respectively, then*

$$T[f_1(w) \star f_2(w)] = \sqrt{\frac{1}{m}} T[f_1(w)] T[f_2(w)], \quad (16)$$

and

$$T[f_1(w)] \star T[f_2(w)] = \sqrt{m} T[f_1(w) f_2(w)]. \quad (17)$$

Proof The proof of this corollary is consistent with the proof of Fourier classical convolution in $L^2(\mathbb{R})$ space, then

$$\begin{aligned} T[f_1(w) \star f_2(w)] &= \sqrt{m} \int_{\mathbb{C}} \int_{\mathbb{C}} f_1(\tau) f_2(w - \tau) d\lambda(\tau) e^{mz\bar{w}} d\lambda(w) \\ &= \int_{\mathbb{C}} f_1(\tau) d\lambda(\tau) [\sqrt{m} \int_{\mathbb{C}} f_2(w - \tau) e^{mz\bar{w}} d\lambda(w)] \\ &= \int_{\mathbb{C}} f_1(\tau) d\lambda(\tau) [\sqrt{m} \int_{\mathbb{C}} f_2(w) e^{mz(\bar{w} + \bar{\tau})} d\lambda(w)] \\ &= T[f_2(w)] \int_{\mathbb{C}} f_1(\tau) e^{mz\bar{\tau}} d\lambda(\tau) \\ &= \sqrt{\frac{1}{m}} T[f_2(w)] [\sqrt{m} \int_{\mathbb{C}} f_1(\tau) e^{mz\bar{\tau}} d\lambda(\tau)] \\ &= \sqrt{\frac{1}{m}} T[f_1(w)] T[f_2(w)], \end{aligned}$$

next, the convolution of the transform domain can be obtained by performing an inverse transform, then

$$\begin{aligned} T^{-1}[T[f_1(z)] \star T[f_2(z)]] &= \sqrt{\frac{1}{m}} \int_{\mathbb{C}} \int_{\mathbb{C}} T[f_1(\mu)] T[f_2(z - \mu)] e^{\frac{1}{m} w \bar{z}} d\lambda(\mu) d\lambda(z) \\ &= \int_{\mathbb{C}} T[f_1(\mu)] d\lambda(\mu) [\sqrt{\frac{1}{m}} \int_{\mathbb{C}} T[f_2(z - \mu)] e^{\frac{1}{m} w \bar{z}} d\lambda(z)] \\ &= \int_{\mathbb{C}} T[f_1(\mu)] d\lambda(\mu) [\sqrt{\frac{1}{m}} \int_{\mathbb{C}} T[f_2(z)] e^{\frac{1}{m} w(\bar{z} + \bar{\mu})} d\lambda(z)] \\ &= f_2(w) \int_{\mathbb{C}} T[f_1(\mu)] e^{\frac{1}{m} w \bar{\mu}} d\lambda(\mu) \\ &= \sqrt{m} f_2(w) [\sqrt{\frac{1}{m}} \int_{\mathbb{C}} T[f_1(\mu)] e^{\frac{1}{m} w \bar{\mu}} d\lambda(\mu)] \\ &= \sqrt{m} f_1(w) f_2(w). \end{aligned}$$

The convolution theorem is divided into time domain convolution theorem and frequency domain convolution theorem, and the two have a dual relationship [14, 19].

However, it is not common to combine the LCT, Fock space, and the convolution theorem as they belong to different fields of mathematics and physics. The LCT is mainly used in signal processing and image processing, while the Fock space is a concept in quantum mechanics [4, 14, 17, 19]. Therefore, although the convolution theorem in the Fock space is not significantly different from that in $L^2(\mathbb{R})$ space in form, there is currently no clear discussion on the physical significance and application background of the convolution theorem of the LCT in the Fock space. If further exploration of the relationship between these three is needed, it may require deeper mathematical and physical knowledge, as well as specific application scenarios and research backgrounds.

The uncertainty principle is an important fundamental principle of quantum mechanics, which states that in a quantum mechanical system, the position and momentum of a particle cannot be determined simultaneously [21–23].

Theorem 4 (Uncertainty principle) *Let the F_α is the FRFT at angle α , F_β is the FRFT at angle β in the FSF, $f(w) \in F^2(\mathbb{C})$, then we have the following inequalities*

$$\|e^{-2i\alpha} f' + zf - ke^{-i\alpha} f\|^2 \|e^{-2i\beta} f' + zf - ne^{-i\beta} f\|^2 \geq 4 \sin^2(\alpha - \beta),$$

Proof The classical uncertainty principle [21] in the FT analysis takes the following form

$$\Delta u_\alpha^2 \Delta u_\beta^2 \geq \frac{\sin^2(\alpha - \beta)}{4}, \quad (18)$$

where $\Delta u_\alpha = \int_{-\infty}^{\infty} |(u_\alpha - k) F_\alpha(u_\alpha)|^2 du_\alpha$, $\Delta u_\beta = \int_{-\infty}^{\infty} |(u_\beta - n) F_\beta(u_\beta)|^2 du_\beta$, k and n are arbitrary real numbers. Therefore, the corresponding norm is in the following form, then

$$\|(u_\alpha - k) F_\alpha\|^2 \|(u_\beta - n) F_\beta\|^2 \geq \frac{\sin^2(\alpha - \beta)}{4}, \quad (19)$$

thus the corresponding result for the FSF is

$$\|(D - k) F_\alpha\|^2 \|(D - n) F_\beta\|^2 \geq \frac{\sin^2(\alpha - \beta)}{4}, \quad (20)$$

where the norm is taken in the FSF, the operator D is given by Theorem 19 in [6], then

$$Df(z) = \frac{1}{2}(f'(z) + zf(z)), \quad (21)$$

recall from Properties 3, that in the paper of the FSF, we have $f_\alpha(z) = f(e^{-i\alpha}z)$, $f_\beta(z) = f(e^{-i\beta}z)$.

Therefore, the corresponding uncertainty principle in the FSF is

$$\begin{aligned}
\|(D-k)F_\alpha\|^2\|(D-n)F_\beta\|^2 &= \|e^{-i\alpha}f'(e^{-i\alpha}z) + zf(e^{-i\alpha}z) - kf(e^{-i\alpha}z)\|^2 \\
&\quad \times \|e^{-i\beta}f'(e^{-i\beta}z) + zf(e^{-i\beta}z) - nf(e^{-i\beta}z)\|^2 \\
&\geq 4\sin^2(\alpha - \beta),
\end{aligned}$$

since the norm in the FSF is rotation invariant, we can rewrite the above as

$$\|e^{-2i\alpha}f' + zf - ke^{-i\alpha}f\|^2\|e^{-2i\beta}f' + zf - ne^{-i\beta}f\|^2 \geq 4\sin^2(\alpha - \beta), \quad (22)$$

we obtain the uncertainty principle of the FRFT in the FSF.

4 Conclusion

The paper examines the effects of the Bargmann transform on classical integral operators such as the LCT, FT, and FRFT in the $L^2(\mathbb{R})$ space. In the FSF, the unitary equivalent forms of these operators sometimes exhibit surprisingly simple structures, occasionally reveal intriguing properties, and often raise natural new questions. The LCBT is capable of generalizing various transforms, making it a valuable tool in fields like quantum information and optical information. For instance, in beam optics, the LCBT is used to describe how laser beams are transformed through optical systems that include lenses, mirrors, and diffraction gratings. The flexibility in selecting the parameter matrix A allows for precise modeling of complex wavefront transformations, thereby improving the design and analysis of optical devices. Furthermore, the simplicity and clarity of the LCBT representation in the FSF are expected to inspire novel approaches in quantum information, particularly in quantum tomography and state reconstruction.

In conclusion, exploring the interaction between the LCBT and fields such as quantum information computation and optical information analysis provides new insights into quantum state transformation, coherent state analysis, and optical information analysis. The study of the LCBT not only enriches the mathematical framework of information analysis but also opens new avenues for research and technological innovation, with wide application prospects.

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Data availability No data was used for the research described in this paper.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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